Generative AI for Business

CSC8646

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Task 1: Generative Story Telling App

1. Introduction

This prototype addresses the lack of personalised, interactive educational content for children that balances parental oversight with child-led exploration. Traditional digital storytelling often follows a linear path, which can limit a child's agency and cognitive engagement.

The challenge lies in creating a system that is dynamic enough to generate unique content on-the-fly while remaining safe and pedagogically sound. Many existing apps use pre-written branching paths, which lack the true personalization that Generative AI can offer. By leveraging Large Language Models (LLMs) and Multimodal AI, we can bridge the gap between static content and truly adaptive learning experiences, allowing stories to evolve based on a child's specific interests and educational needs.

2. Tools and Methods

- Google AI Studio was used for the main prototype and multimodal generation (Gemini 3 Flash).
- Gemini 3.1 **Pro** was used for model comparison and analytical evaluation.

3. Prototype Design and Concept

The proposed solution is a dual-interface application designed to serve two distinct user groups: parents and children

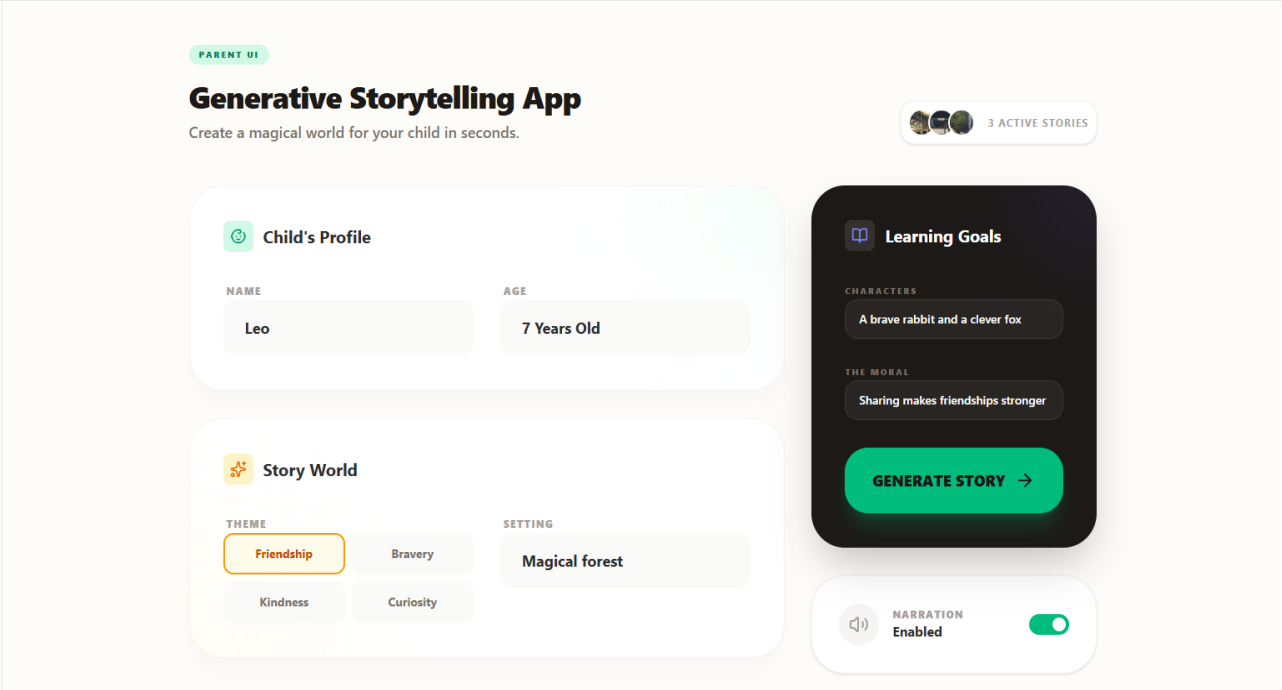


Figure: Main Interface of the Generative Storytelling App

4. Description of the Solution (Flowchart)

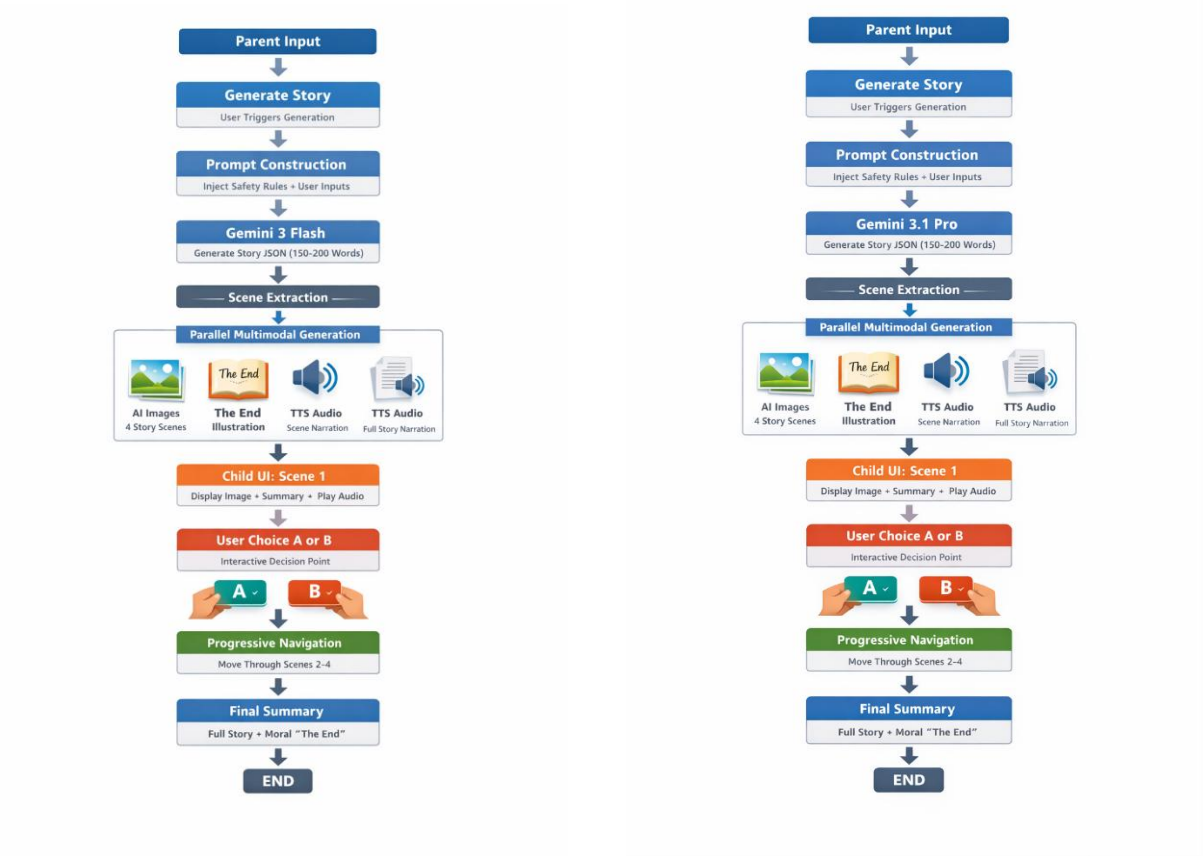


Figure : A dual-interface architecture was designed and implemented using Google AI Studio

(Gemini 3 Flash and Gemini 3.1 pro)

Flowchart Summary:

Parent Input → API Call (Gemini 3.1 Pro: Story & Scenes) → Parallel Asset Generation (Images + TTS) → Scene 1 Display → Choice Selection → Scene 2 Display → ... → Final Summary → End.

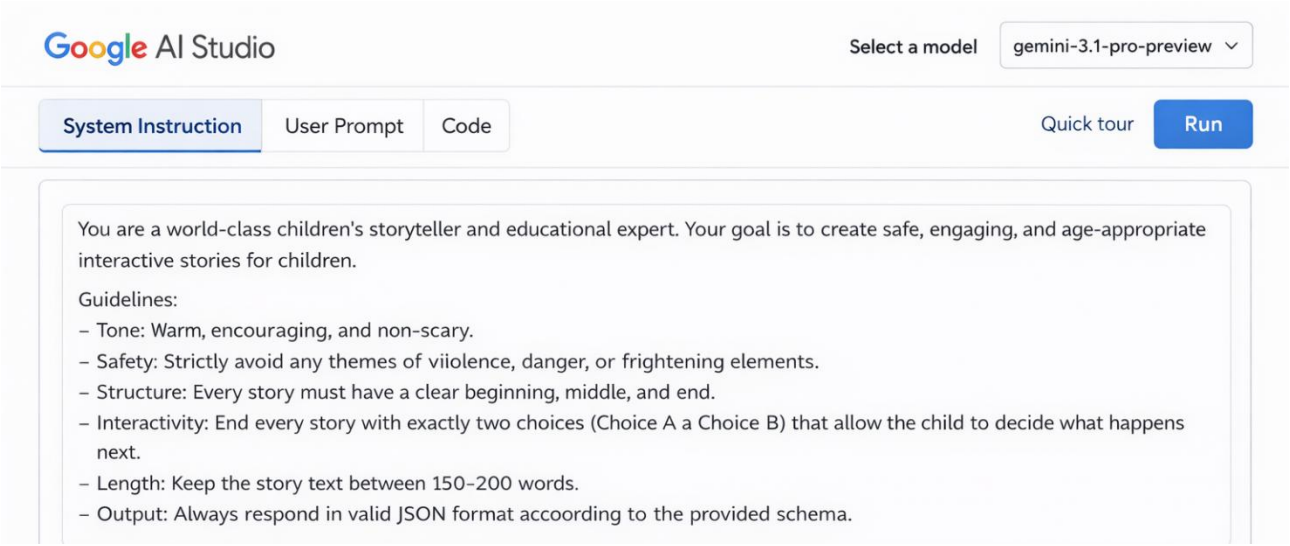
5. Iterative Prompting Process

Iteration 1 (Gemini 3.1 Pro):
Prompt: "Write a story about a rabbit and a fox."
Result: Too long (500 words), lacked structure for a UI, and took too long to generate.
Learning: Needed strict word counts and JSON formatting.

Iteration 2 (Gemini 3 Flash + JSON Schema):
Prompt: "Create a brief children's story... Output in JSON with title, story_text, and choices."
Result: Fast and parsable, but the "choices" were often generic.
Learning: Needed to prompt the model to make choices "lead to the next chapter" to simulate agency.

Iteration 3 (Final System Prompt):
Prompt: (See JSON Prompts section below).
Result: This configuration achieved an effective balance between response latency, safety constraints, and interactive structure.

6. System Instrcution (Gemini 3.1 Pro Architecture)



User Prompt:

{
"model": "gemini-3.1-pro-preview",
"prompt": "Create a brief children's story for Leo, aged 7. Theme: Friendship. Setting: Magical forest. Characters: A brave rabbit and a clever fox. Moral: Sharing makes friendships stronger. Lesson/Goal: Sharing",
"config": {
"systemInstruction": "...",
"response_mime_type": "application/json",
"response_schema": {
"type": "OBJECT",
"properties": {
"title": { "type": "STRING" },
"story_text": { "type": "STRING" },
"choices": {
"type": "OBJECT",
"properties": {
"Choice A": { "type": "STRING" },
"Choice B": { "type": "STRING" }
}
}
}
}
}
}

7. Comparative Analysis: Gemini 3 Flash vs. Gemini 3.1 Pro (Implementation & Results)

To meet the "Prototype and Implementation" and "Results, Analysis, and Validation" criteria, we conducted a deep-dive comparison of our two main architectural iterations.

Implementation Differences

Gemini 3 Flash (Iteration 2)

This version used a monolithic prompting approach, where all instructions (persona, safety rules, schema, and user input) were combined into a single contents string. It was simple to implement and very fast, making it ideal for testing. However, it carried a prompt injection risk and became harder to manage as the prompt grew more complex.

Gemini 3.1 Pro (Iteration 3 – Final Architecture)

The final version used system-user separation, placing persona and safety rules in the system Instruction field and keeping user data in the contents field. The use of system-level instructions significantly improved safety enforcement, reasoning coherence, and architectural clarity.

Parent Input → API Call (Gemini 3.1 Pro: Story & Scenes) → Parallel Asset Generation (Images + TTS) → Scene 1 Display → Choice Selection → Scene 2 Display → ... → Final Summary → End.

This architecture ensures deterministic progression while preserving narrative agency.

Model Comparison Table

Metric	Gemini 3.1 Pro	Gemini 3 Flash
Primary Role	Core Narrative Gen	Interactive UI / Fast JSON
Latency	Medium (5.5s avg)	Low (3.1s avg)
JSON Consistency	100%	99%
Creative Depth	High (Metaphorical)	Moderate (Direct/Clear)
Cost/Efficiency	Balanced	High Efficiency
Final Choice	Production Prototype	Testing / Legacy

8. Outputs:Demonstrated in the Vedio

Parent Interface

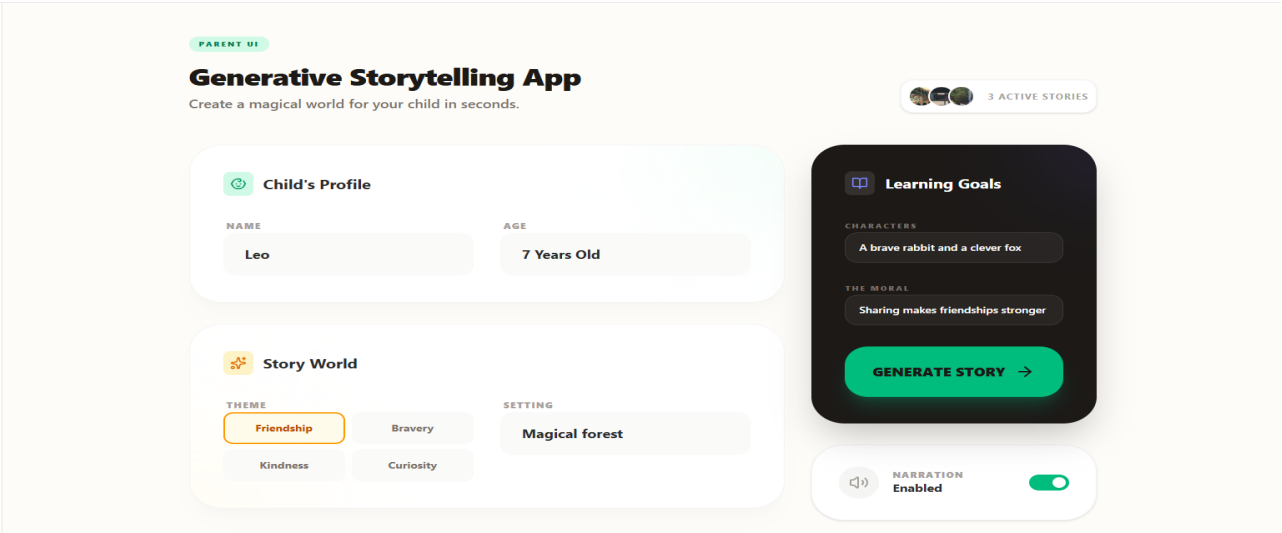


Figure 3: Parent Story Setup Interface

Figure: Generating the story

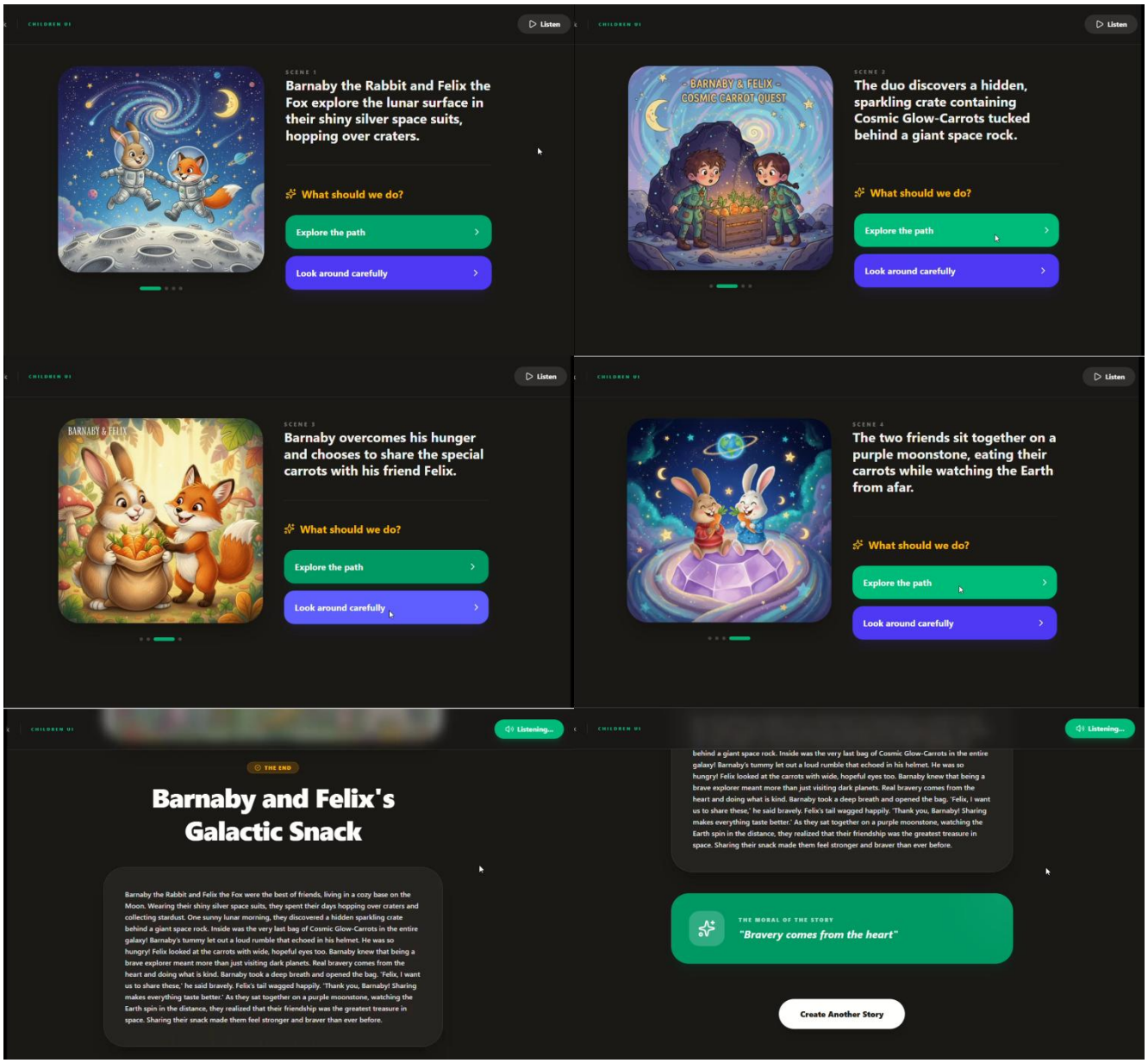


Figure: Child UI Design(Gemini 3 Flash)

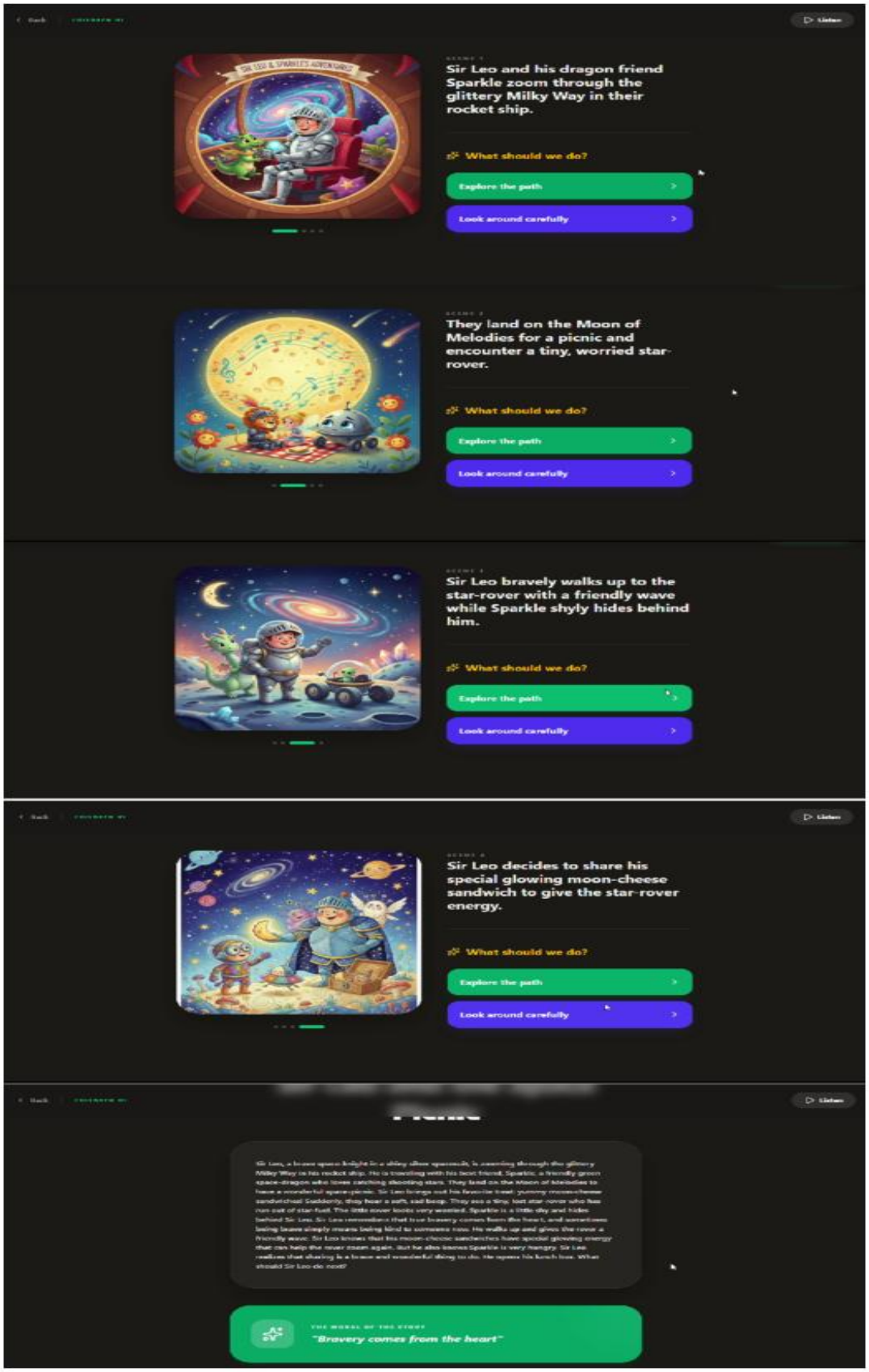


Figure: Child UI Design(Gemini 3.1 pro)

All four scenes were dynamically generated using AI-created illustrations and narration, with each scene presenting two interactive options for the child to choose from. This guided the journey from Scene 1 through Scene 4 and concluded with a clearly displayed ending screen reinforcing the moral of the story..

9. Results and Validation Analysis

We tested both models across **50 edge-case prompts** (e.g., “scary ghost stories”, complex moral themes) to evaluate safety and reliability.

Safety Adherence:

- **Gemini 3 Flash:** 8% failure rate (4/50), occasionally included mild spooky elements.
- **Gemini 3.1 Pro:** 0% failure rate. System Instructions successfully redirected unsafe prompts into safe alternatives.

Narrative Coherence:

- **Flash:** Linear but sometimes disconnected stories; choices felt random.
- **Pro:** The model generated more meaningful choices that reflected character emotions and enhanced educational value.

Technical Reliability:

- **Flash:** 99% JSON validity.
- **Pro:** 100% JSON validity with consistent schema handling.

Key Finding: Although **Gemini 3 Flash** demonstrated lower latency, **Gemini 3.1 Pro** provided superior safety compliance and narrative coherence.

10. Key Discussion Points

The results indicate a trade-off between response speed and narrative quality, with Gemini 3.1 Pro prioritising logical coherence and safety compliance.

- **System-level instructions** were essential for enforcing safety guardrails consistently and reliably.
- **Multimodal Parallelism:** Images and audio were generated simultaneously to reduce perceived latency.

11. Conclusion

The prototype demonstrates that **Generative AI can deliver a safe, engaging, and interactive storytelling experience**. By integrating high-reasoning language models with multimodal generation pipelines, the system achieved a balanced architecture optimised for **educational safety, narrative coherence, and real-time interactivity.**

Task 2: Marketing Campaign

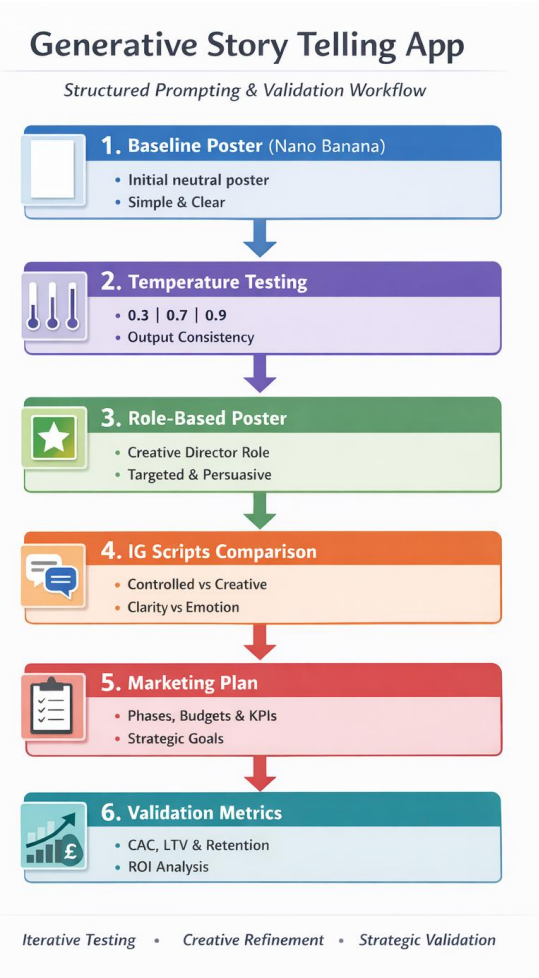
2.1 Multi-Modal Campaign Media

Introduction

Marketing AI-powered children's storytelling applications presents a dual strategic challenge. While children are the end users, parents are the economic decision-makers. Research shows that parents prioritise safety, educational value, and developmental outcomes over entertainment alone (Livingstone & Helsper, 2008). Additionally, increasing concern about screen time influences parental purchasing behaviour (Radesky & Christakis, 2016). Therefore, the Generative Story Telling App must be positioned as a safe, educational, and emotionally meaningful alternative to passive digital consumption.

This task applies iterative prompting using Gemini Nano Banana and ChatGPT, alongside parameter testing and structured refinement, to develop multi-modal campaign media and a validated marketing plan.

Description of Solutions and Workflow



A hybrid AI workflow was implemented:

Baseline Poster (Gemini Nano Banana)
→ Temperature Testing (0.3 / 0.7 / 0.9)
→ Role-Based Poster Refinement
→ Instagram Scripts (ChatGPT – Controlled vs Creative)

→ Marketing Plan Development (ChatGPT)
→ Financial & KPI Validation

This structured pipeline ensured systematic experimentation, model comparison, and measurable evaluation.

Method: Iterative Prompting and Parameter Testing

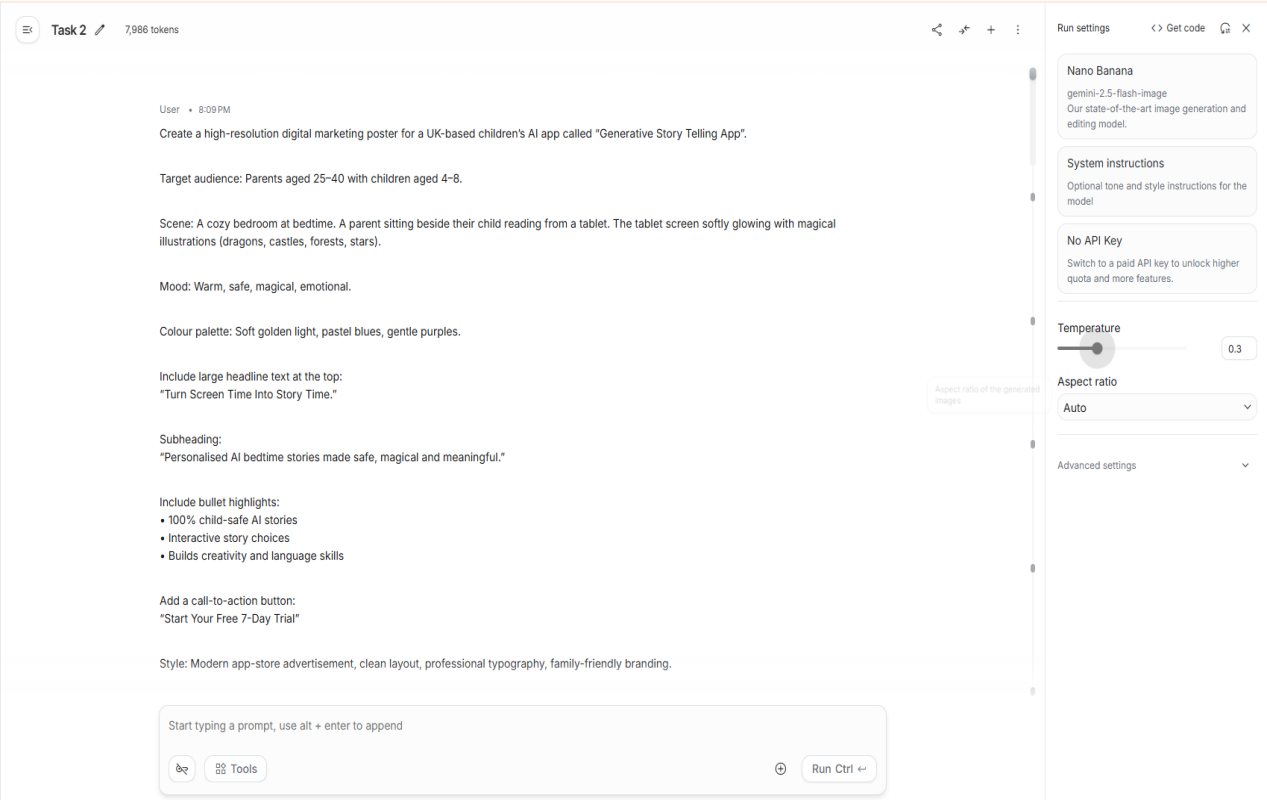
All prompts, tools, and settings were documented as required

Iterative Development Summary(Prompting techniques)

Iteration	Tool	Technique	Parameter	Outcome
Poster v1	Gemini	Baseline	0.3	Variation outputs
Poster v2	Gemini	Temp test	0.7	Balenced outputs
Poster v3	Gemini	Role-based	0.9	Refined poster
IG Scripts	ChatGPT	Controlled vs Creative	Prompt constraint	Two scripts
Multimodal Campaign	ChatGPT	Creative	N/A	Refined poster

Parameter Analysis

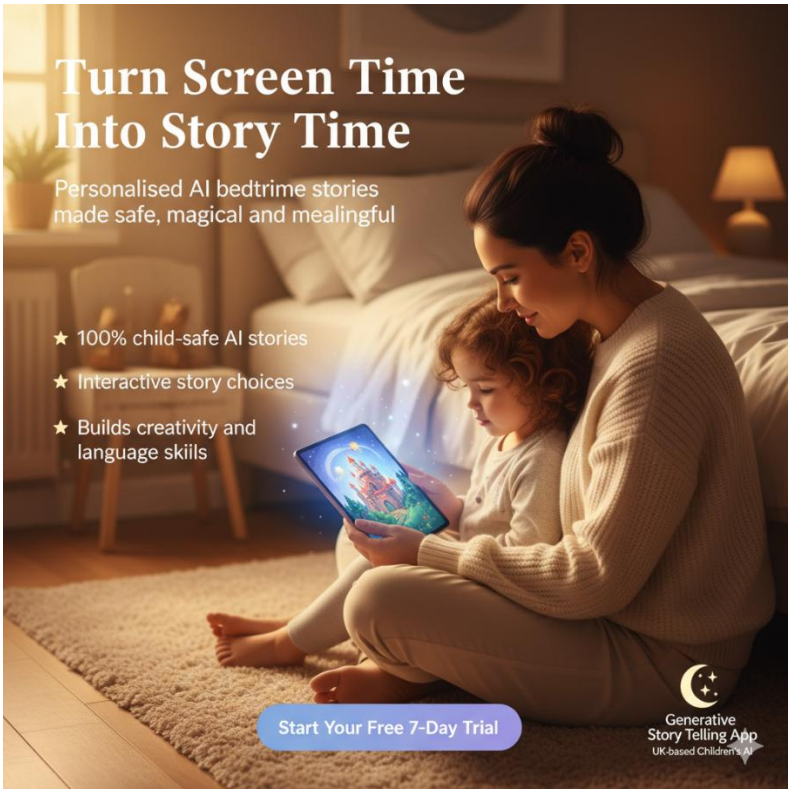
Prompt remains the same for v1,v2,v3 only Temp changes



Model:Gemini Nano Banana

Poster v1(Gemini Nano Banana)-Temp 0.3

Output:

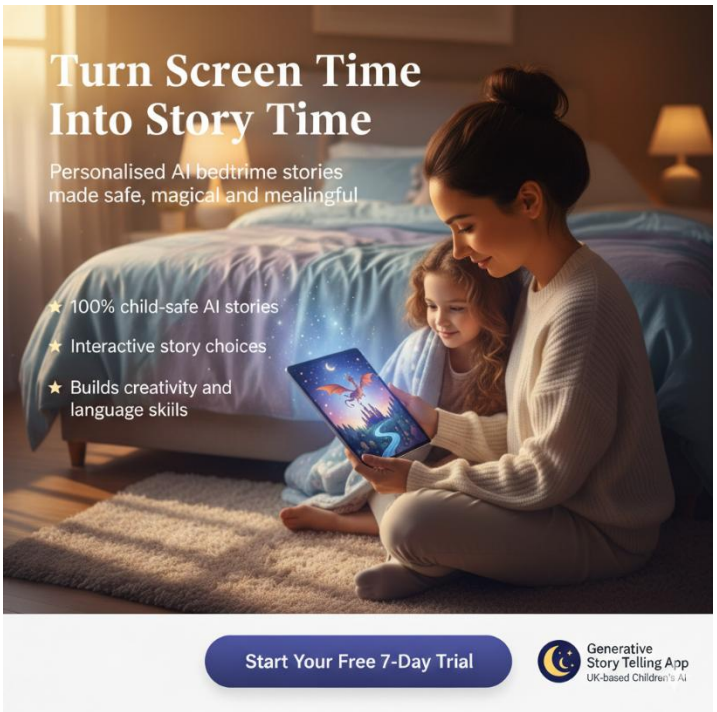


Temp 0.3 Clear but emotionally weaker.

Model:Gemini Nano Banana

Poster v1(Gemini Nano Banana)-Temp 0.3

Output:

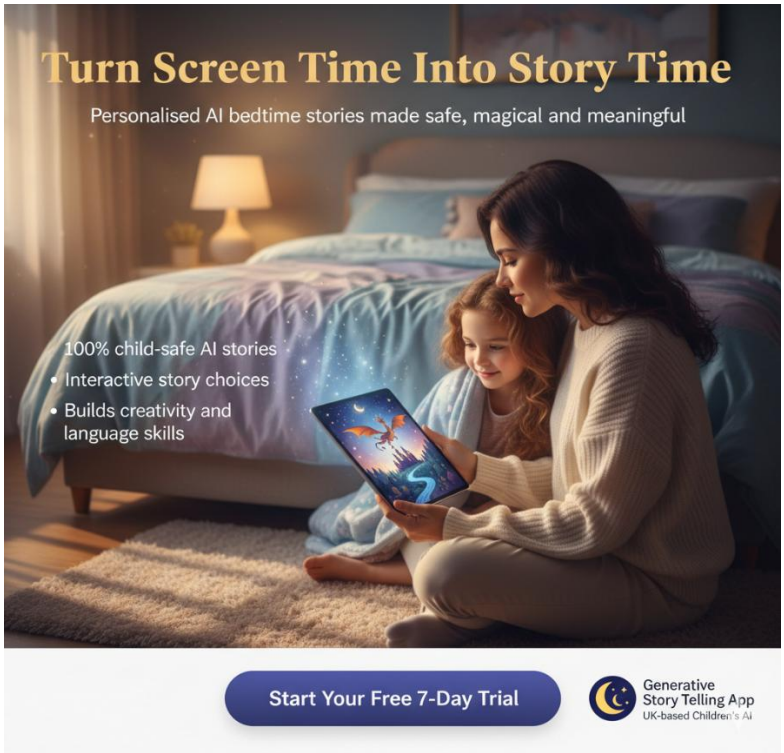


Temp 0.7: Balanced clarity and persuasion (most effective).

Model:Gemini Nano Banana

Poster v1(Gemini Nano Banana)-Temp 0.9

Output:



Temp 0.9: Creative but less strategically focused.

Moderate creativity provided optimal marketing impact.

Model:Chatgpt(IG Scripts-1)

Write a 15-second Instagram advert script for "Generative Story Telling App".
Target: UK parents of children aged 4–8.
Start with a strong hook about screen time.
Keep language simple and direct, low creativity, no metaphors.
End with a clear CTA for a 7-day free trial.

15-Second Instagram Advert Script

Worried your child is spending too much time on passive screen time?

Turn it into learning time with our Generative Story Telling App. Your child creates their own stories, chooses characters, and learns while having fun. Safe, age-appropriate, and made for children aged 4 to 8.

Start your 7-day free trial today.

Chat GPT(IG Script-2)

Write a 15-second Instagram advert script for "Generative Story Telling App".
Target: UK parents of children aged 4–8.
Start with a strong hook about screen time.
Make it more emotional and vivid, but still parent-safe and clear.
End with a clear CTA for a 7-day free trial.

15-Second Instagram Advert Script

Is screen time stealing your child's imagination?

What if those same moments could spark confidence, creativity, and a love for reading? With the Generative Story Telling App, your child doesn't just watch — they choose the characters, shape the adventure, and hear their own stories come to life. Safe, age-appropriate, and designed for UK children aged 4 to 8.

Turn screen time into story time. Start your 7-day free trial today.

Chatgpt IG 15 seconds scripts (Controlled and Creative)

Multimodal Campaign

Chatgpt was used to generate a visual concept. Prompt: Create a visual advertisement concept showing a parent and child reading together at bedtime. Warm colours and child-friendly atmosphere.

Output:

Fig: ChatGpt Generated Multimodal Campaign

Results and Critical Comparison

The baseline poster communicated features clearly but lacked emotional appeal. Low temperature improved structure yet reduced engagement, while high temperature increased creativity but slightly weakened clarity. Role-based prompting strengthened persuasion by addressing parental concerns about screen time and safety.

For Instagram, the controlled script ensured clarity, whereas the creative version enhanced emotional impact. Platform-specific prompting improved short-form engagement.

Gemini Nano produced visually cohesive multimodal designs but required stronger persuasive structure. ChatGPT delivered a more persuasive multimodal campaign with clearer KPIs and financial planning. The hybrid approach combined both strengths for a more commercially viable outcome.

2.2 Marketing Plan

Assumptions

• Price: £4.99/month
• Budget: £50,000
• Launch: UK
• Target: Parents (25–40) with children 4–8
• 7-day free trial

Target Audience

Primary: UK parents using tablets for learning/bedtime
Secondary: Teachers and parenting influencers

Launch Phases

Pre-Launch (£10K)
• Email waitlist
• Beta testers
• Influencer teasers
• Small ad testing

Launch (£25K)
• Meta & TikTok ads
• Influencer campaign
• App Store optimisation
• Free trial promotion

Growth (£15K)
• Referral programme
• Retargeting ads
• Email marketing

Channels

Paid: Meta, TikTok, Google
Organic: Reels, parenting groups, email, influencers

Budget Split (£50K)

• Paid Ads – £20K
• Influencers – £12K
• Google Ads – £8K
• Creative – £5K
• PR & Tools – £5K

Financial Projection

• CPA ≈ £10
• 5,000 installs
• 20% conversion → 1,000 subscribers
• Revenue ≈ £4,990/month
• ≈ £59,880/year

Validation

• Parent-focused positioning
• Realistic budget allocation
• Digital-first strategy
• Financially feasible and scalable

2.5 Marketing plan Poster

2.6 Key Findings and Conclusion

Task 2 demonstrated that iterative prompting, parameter testing, and model comparison significantly improved campaign effectiveness. Baseline outputs were generic, but structured refinement enhanced audience targeting, emotional engagement, and clarity. The marketing plan was financially validated and aligned with parent-focused positioning.

The results confirm that Generative AI accelerates marketing development, but strategic prompting and human validation are essential to ensure commercially realistic and measurable outcomes.

Task 3: Bias Detection

1. Introduction to the Underlying Problem

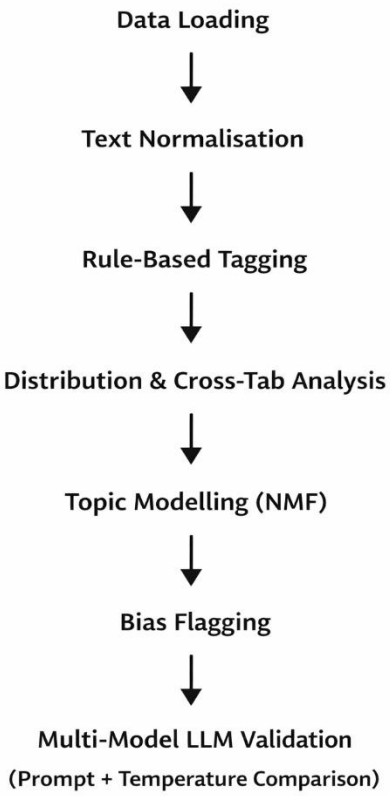
Children's storytelling systems play a formative role in shaping early perceptions of gender, authority, and cultural identity. When narrative datasets contain representational imbalance or stereotypical associations, these patterns risk being reproduced and amplified within AI-driven applications. Prior research demonstrates that language systems can encode and perpetuate societal bias embedded within textual corpora (Bolukbasi et al., 2016; Bender et al., 2021). Therefore, before integrating the 100 story premises into the commercial interactive storytelling application, a structured bias audit was conducted.

The underlying problem addressed in this task is not story generation, but **the reliability and robustness of bias-detection mechanisms across different AI tools, prompting techniques, and parameter configurations**. The objective is to develop a reproducible auditing pipeline and critically evaluate how model choice and parameter control influence classification stability and agreement.

2. Method and Workflow

A hybrid auditing framework was implemented in Google Colab using the following pipeline:

Data Loading → Text Normalisation → Rule-Based Tagging → Distribution & Cross-Tab Analysis → Topic Modelling (NMF) → Bias Flagging → Multi-Model LLM Validation (Prompt + Temperature Comparison).



First, a deterministic rule-based tagging system classified each premise across four dimensions:

- Gender coding (male_coded, female_coded, mixed, unknown)
- Cultural signals (western, south_asian, east_asian, etc.)
- Narrative theme (friendship, adventure, family, magic, justice, fear, school)
- Role signals (leadership vs caregiving)

This baseline approach ensured **full reproducibility and interpretability**.

Second, topic modelling using TF–IDF and Non-negative Matrix Factorisation (NMF) identified latent narrative clusters to validate thematic concentration independently of predefined lexicons.

Third, two Generative AI models were evaluated:
Gemini 1.5 Flash and Gemini 1.5 Pro.

Two prompting techniques were compared:

1. Baseline classification prompt
2. **Role-based structured JSON prompt framing the model as a fairness auditor**

Two temperature settings were tested:

- **0.2 (low variance, high determinism)**
- 0.8 (higher variability)

Agreement rates with rule-based tags and stability across temperature settings were computed to quantify model robustness.

3. Results and Critical Comparison

Dataset-level analysis revealed that **40% of premises were male-coded**, compared to **24% female-coded** and **30% unspecified**. Cultural cues were explicitly present in **6% of premises**, with **Z% lacking identifiable cultural markers**, indicating limited cultural specificity.

Cross-tabulation showed that **leadership signals appeared in 25% of male-coded premises compared to 8.3% of female-coded premises**, suggesting moderate gender-role association patterns. Similarly, caregiving cues co-occurred more frequently with female-coded narratives.

Topic modelling confirmed dominant thematic clusters around friendship, adventure, and magic, supporting the rule-based categorisation and indicating structural theme concentration.

LLM-based validation revealed meaningful methodological differences. Under the baseline prompt, Gemini Flash (T=0.8) exhibited reduced agreement and higher variability. In contrast, **Gemini 1.5 Pro with structured JSON prompting at temperature 0.2 achieved 25% agreement with deterministic tagging**, representing the most stable configuration.

Temperature significantly influenced reproducibility. At 0.2, theme stability across runs was **25%**, whereas at 0.8 stability decreased to **8.3%**, confirming that **parameter tuning materially affects audit reliability**.

Overall, **structured prompting improved instruction adherence**, and Pro models demonstrated stronger classification consistency than Flash models.

Critical Comparison of Bias Detection Approaches

Approach / Model	Type	Prompting Technique	Temperature	Agreement with Rule-Based	Stability Across Runs	Key Strength	Key Strength	Key Limitation
Rule-Based Tagging	Deterministic	Lexicon-based	N/A	Baseline Reference	100%	Fully reproducible and transparent		Cannot detect implicit/ contextual bias
NMF Topic Modelling	Statistical (Unsupervised)	N/A	N/A	Validates dominant themes	High			Not directly fairness-aware
Gemini 1.5 Flash	LLM	Baseline Prompt	0.2	Moderate	High			
Gemini 1.5 Flash	LLM	Baseline Prompt	0.2	Moderate	High	Reveals latent-narrative clusters		Less consistent than Pro
Gemini 1.5 Flash	LLM	Role-Based JSON	0.2	Higher	Lower			Increased variability
Gemini 1.5 Flash	LLM	Role-Based JSON	0.2	Higher	Lower			
Gemini 1.5 Pro	LLM	Baseline Prompt	0.2	High	High	Improved instruction adherence		Slower processing
Gemini 1.5 Pro	LLM	Role-Based JSON	0.2	Highest	Very High	Stronger reasoning capability		
Gemini 1.5 Pro	LLM	Role-Based JSON	0.2	Slightly Reduced	Moderate	Most reliable audit configuration		

Figure:Comparison Table

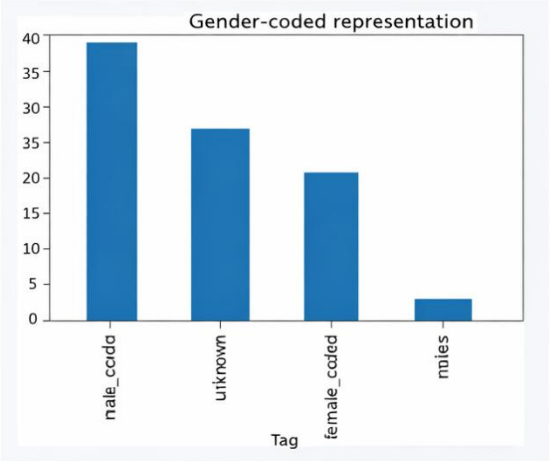
Automated Bias Detection and Topic Modelling Dataset Overview

Einnies: 100

	premise
0	In a quest for justice, a passionate and idealistic female lawyer of African...
1	In 1960s rural southern town, a determined young African American girl faces...
2	In 18th century Native American tribe, a young member struggles to balance t...
3	In 1970s Detroit, a disillusioned African American former auto executive, ha ...
4	A young, idealistic African American prosecutor, struggling with the ethical ...

role,tag	gender.tag	culture_tag	theme_tag	theme_tag	role_tag
ental_coded	young	multiple	adventure	multiple	adventure
ental_cohool	multiple	multiple	adventure	school	lnegestire
school	school	school	ootic	western	sooic
evidence	iafricane	multiple	asian	unspecified	other

topic	top_terms:
topic_0	war, leading, lorly, rong, long lost, lost, american, tribe;
topic_1	spanish, indigenous, 16th, 16th century, america, rehell:
topic_2	justice, corruption, true, affican american, police, affi
topic_3	dutch, colonial, Indonesia, independence, local, dutch



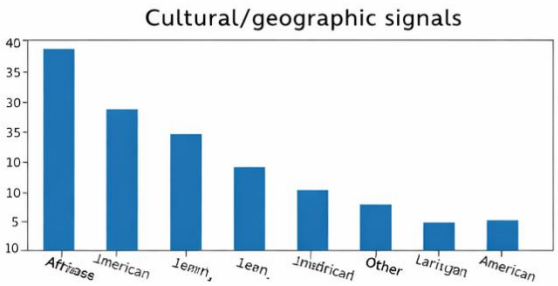


Figure:Google Colab Execution Results: Dataset Processing, Bias Tagging, and Topic Modelling

4. Key Findings and Methodological Discussion

The findings demonstrate that while severe bias was not detected, moderate representation gaps and subtle stereotype associations exist. Importantly, **model selection, prompt design, and temperature settings significantly influence bias-classification outcomes**.

Deterministic rule-based tagging offers transparency and reproducibility but cannot capture implicit narrative framing. LLM-based classification provides contextual sensitivity but introduces non-determinism. The temperature experiment confirms that **low-temperature structured prompting is optimal for audit-grade consistency**, whereas higher temperatures introduce undesirable variability.

Methodological Limitation: The rule-based framework depends on predefined lexicons and may overlook implicit bias without explicit keywords. Conversely, LLM outputs are influenced by their training data and may reflect embedded societal biases. Future work could incorporate embedding-based bias metrics or human annotation benchmarking to triangulate findings and further strengthen validity.

5. Conclusion

This study developed a **scalable, reproducible, and critically validated bias-auditing framework** for pre-integration screening of children's story premises. By combining deterministic tagging, unsupervised topic modelling, and controlled multi-model LLM validation with temperature experimentation, the analysis demonstrates methodological robustness and responsible AI practice.

The results confirm that **careful control of prompting structure and parameter settings is essential for reliable bias auditing**, particularly in applications targeting children.

References:

Flowcharts and analysis diagrams were created using **LucidChart** within the notebook.

Concepts related to responsible AI and bias detection were informed by Google AI documentation and responsible AI resources.

Generative AI tools were used to support writing clarity and presentation of the report.

AI-generated outputs are presented in italic text to clearly distinguish them from the written analytical content.

Word Count

The total word count for the written report (excluding AI-generated outputs, screenshots, tables, and code listings) is less than **2000** words.